

Tier 1: Macro Plateau Controller

π_{macro}

[Semi-MDP, $\Delta_{\text{seg}} = 100$]

- **Observation** $s_k^{\text{macro}} \in \mathbb{R}^{13}$: Stagnation rate, temp T/T_0 , route separation rb , capacity balance lb , tight-TW ratio τ_{tight} , slack $\bar{s}$, pool occupancy ρ_{pool}
- **Network**: Dueling Double DQN ($13 \rightarrow 128 \rightarrow 128 \rightarrow 7$) with Prioritized Replay
- **7 Strategic Regimes** (a_0 – a_6): a_0 : Default exploration; a_1 : Intensify / deep cool; a_2 : Thermal reheat; a_3 : TW-Rescue; a_4 : GEC route elimination; a_5 : Set-Part Recomb; a_6 : Infeasible descent (adaptive TW/cap penalty relaxation)

Mode a_5 Trigger

Mode a_k (every Δ_{seg} iters)

Tier 2: Micro Selection Tier (π_{micro} & LAC)

[Step $t \in \{1, \dots, T\}$]

- **Observation** $s_t^{\text{micro}} \in \mathbb{R}^{20}$: Operator success history, removal ratios, customer spatial clustering, normalized forward time-slack distribution
- **Operator Controller**: Dueling DDQN ($20 \rightarrow 128 \rightarrow 65$) selecting 65 action pairs (13 Destroy $\times$ 5 Repair) with Softmax-Entropy confidence gate $w_{\text{conf}}(s)$
- **Learned Acceptance (LAC)**: MLP ($9 \rightarrow 64 \rightarrow 48 \rightarrow 32 \rightarrow 1$) evaluates delayed downstream gain ($H_{\text{lac}} = 80$ lookahead); replaces SA after 300 warmup steps

Operator pair (d, r) (iter t)

Tier 3: Heuristic Execution Engine (ALNS)

[Execution Muscle]

- **Composite Operators**: 13 Ruin (Shaw, Worst, Slack, GEC, Cluster) $\times$ 5 Recreate (Greedy, Regret-2/3/4, Deep-Insertion)
- **Spatial Guidance**: 3-layer Contrastive GNN ($k = 25$ kNN) prunes 98.74% edges
- **Fast Feasibility**: $O(1)$ Savelsbergh forward time-slack evaluation
- **Fleet Elimination**: Polynomial Generalized Ejection Chains (GEC)

Feasible s_{cand}

Dual-Slot Route Pool $\mathcal{P}$ [Cap $|\mathcal{P}|_{\text{max}} \leq 2000$]

- **Deduplication**: Frozen set customer hash table guarantees 0 duplicate columns
- **Slot A (75% Quota)**: Cost-per-stop efficiency metric $c_r/|r|$
- **Slot B (25% Quota)**: Short-route fragments for vehicle reduction
- **Eviction Policy**: FIFO displacement upon reaching capacity boundary

Candidate columns in $\mathcal{P}$

Set Partitioning Recombination (HiGHS)

[Exact Recomb]

- **MILP Model**: $\min \sum_{r \in \mathcal{P}} c_r x_r + \lambda_{\text{pen}} \sum_i y_i$ s.t. $\sum_{r \ni i} x_r + y_i = 1$
- **Lexicographic Penalty**: $\lambda_{\text{pen}} \in \{2 \dots 50\} \bar{c}$ guarantees exact $N_V \succ D_T$
- **Cutoff**: Deterministic 4.0s wall-clock MIP limit; returns s_{inc} if timeout

Recombined solution s_{MIP}

Incumbent Plan s_{inc} & Elite Archive

[Feasibility Gate]

- **Feasibility Invariant**: 100% hard capacity and time-window conformance
- **Lexicographic Admission**: $s_{\text{cand}} \prec s_{\text{inc}} \iff N_V^{\text{cand}} < N_V^{\text{inc}} \vee (N_V^{\text{cand}} = N_V^{\text{inc}} \wedge D_T^{\text{cand}} < D_T^{\text{inc}})$
- **Cold-Start Protocol**: Isolated ephemeral scratch directories (0 seed leakage)

Elite routes $\rightarrow \mathcal{P}$